

Technology #162

Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-09

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without soldering; apply a thin hydrophobic or low-surface-energy coating (e.g., PTFE or silane-based) to the coil exterior and shroud interior to reduce scale adhesion, and add a short straight inlet diffuser to stabilize flow before the helical section.

WHY DOES IT MATTER?

Documented problem 'scale deposition and pitch deformation in helical coil' concerning countercurrent heat recovery module for espresso machine. User-submitted engineering problem (session input): "A compact countercurrent heat recovery module for commercial espresso machines: clean incoming water is preheated by waste heat from the steam boiler exhaust using a coiled helical copper tube wound at uniform pitch inside a cylindrical stainless steel shroud, with flanged inlet and outlet nozzles a". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 20 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 8-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

constraint-compliance envelope article

WHAT WOULD MAKE US KILL IT?

killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 8-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: constraint-compliance envelope article

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - constraint-compliance envelope article - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: constraint-compliance characterization. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-002 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling / kink margin tests across duty cycle. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-004 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling/kink margin test across the duty cycle and path geometry (measured quantity: mechanical stress / fatigue endurance). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: constraint-compliance envelope article. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: The helical coil is treated as a self-supporting structural spring whose pitch stability comes from elastic preload and distributed mechanical constraint, not from metallurgical bonding; scale deposition is reduced by maintaining a continuous thin-film flow regime and by making the coil surface energetically unfavorable to calcium carbonate nucleation.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'scale deposition and pitch deformation in helical coil' concerning countercurrent heat recovery module for espresso machine. User-submitted engineering problem (session input): "A compact countercurrent heat recovery module for commercial espresso machines: clean incoming water is preheated by waste heat from the steam boiler exhaust using a coiled helical copper tube wound at uniform pitch inside a cylindrical stainless steel shroud, with flanged inlet and outlet nozzles a". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as

absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without soldering; apply a thin hydrophobic or low-surface-energy coating (e.g., PTFE or silane-based) to the coil exterior and shroud interior to reduce scale adhesion, and add a short straight inlet diffuser to stabilize flow before the helical section.

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without soldering; apply a thin hydrophobic or low-surface-energy coating (e.g., PTFE or silane-based) to the coil exterior and shroud interior to reduce scale adhesion, and add a short straight inlet diffuser to stabilize flow before the helical section.

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: epsilon.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.

- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'scale deposition and pitch deformation in helical coil', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	coil must hold pitch without solder joints and survive daily descaling cycles	SOURCE_FACT
DI-002	Mechanism driver (proposed)	The helical coil is treated as a self-supporting structural spring whose pitch is modified from elastic preload and distributed	MODIFIED
DI-003	Expected effect target	Coil pitch remains within $\pm 2\%$ of nominal after 500 daily descaling cycles and modified thermal expansion; scale mass a	MODIFIED
DI-004	Domain design input pattern: delivery path applicability and NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	
DI-005	Domain design input pattern: actuation applicability and NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	
DI-006	Domain design input pattern: duty cycle applicability and NOT_ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED	
DI-007	Regulatory design input: compliance candidate standards (ASTM F2516 (ratrod) vs. pathway UNKNOWN (PERFECT PROPOSED))	is never inferred	
DI-008	Custodied observation: Design of coiled helical coil heat exchanger 2015-09-28; content_hash 41c96ca1539b31ce376a39f89ef962b8f736d92e0c		
DI-009	Custodied observation: Design of high pressure reactor system for lab 2015-09-28; content_hash 41c96ca1539b31ce376a39f89ef962b8f736d92e0c		
DI-010	Custodied observation: TRANSISTOR AND POWER-HYBRID C-ON-ONE-ESSENTIALS FOR THE DESIGN OF		
DI-011	Custodied observation: CFD simulation of helical coil with Diffusion 2.0; content_hash 25be790b99ad52106392c849a25cd		
DI-012	Custodied observation: APPLICATIONS OF MACHINE LEARNING TO NON-DESTRUCTIVE TESTING OF		
DI-013	Custodied observation: Specification of a coil (0.043x0.006x0.17u; content_hash 66ff8a50230fb8063026aef15a2522708c		
DI-014	Custodied observation: MEME: An Accurate Modeling Method 1966; content_hash 8026a1883ca935a69a363116f7		
DI-015	Custodied observation: Developed a coil 10x10x10 scale; content_hash 60b91c0e10b24f28a5a11788b91f1e5ee1f6b15b5ee		
DI-016	Custodied observation: The dimensionality of a coiled helical coil arxiv.2407.02493; content_hash 1fa2e55499206b9576a43776b20f8525b0f4b458fa3d		
DI-017	Custodied observation: An Investigation of a Coiled Helical Coil Heat Transfer 1974; content_hash 11819d2245435cb85c6e22948ffd900f677d		
DI-018	Custodied observation: Thermal Source Modeling of a Coiled Helical Coil 5361692d2947aa7cd87ce816d4e47fd7137047ee5fa30118ca5eb		
DI-018	Tool Feed Drive System using FEA Method		
DI-019	Custodied observation: Heat Integrated Milk Powder Production 07be0c34b44b84a005f3b536e134da03100e5906ec709ac5183f6aa		
DI-020	Custodied observation: Counter-electroning 3702461848a16921d4f05f96ba9a526a635c986207ac0e6a5f19208d		

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Wind the copper tube on a mandrel with a coiled helical preloaded	CONFIRMED	the coil is slightly compressed against the
DO-002	actuation or compliance element realizing Wind the copper tube on a mandrel with a coiled helical preloaded	CONFIRMED	the coil is slightly compressed against the
DO-003	guidance / bearing interfaces realizing Wind the copper tube on a mandrel with a coiled helical preloaded	CONFIRMED	the coil is slightly compressed against the cy
DO-004	fatigue-critical features register realizing Wind the copper tube on a mandrel with a coiled helical preloaded	CONFIRMED	the coil is slightly compressed against the
DO-005	load path / structural backbone interfaces with actuation of a compliance element: signal energy and force (dimensions: wind the copper tube on a mandrel with a coiled helical preloaded)	CONFIRMED	
DO-006	actuation or compliance element interfaces with guidance of a compliance element: signal energy and force (dimensions: wind the copper tube on a mandrel with a coiled helical preloaded)	CONFIRMED	
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal energy and force (dimensions: wind the copper tube on a mandrel with a coiled helical preloaded)	CONFIRMED	
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by the device, see definitions	CONFIRMED	a mandrel with a controlled
DO-009	Flow/force/signal path geometry realizing the mechanism of a coiled helical preloaded	UNKNOWN	the coil is slightly compressed
DO-010	Operating range for member slenderness ratio within which the mechanism holds its stated behavior	UNKNOWN	bounds for member slenderness ratio; verification
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its stated behavior	UNKNOWN	computed bounds for yield/ultimate strength; verification of ra

ID	Description	Status	Missing Inputs
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its state at a given time	UNKNOWN	Does its state depend on time; verification of range
DO-013	Operating range for required run-out cycles within which the mechanism holds its state at a given time	UNKNOWN	Does its state depend on time; verification of range
DO-014	Operating range for articulation radius within which the mechanism holds its state at a given time	UNKNOWN	Does its state depend on time; verification of range
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its state at a given time	UNKNOWN	Does its state depend on time; verification of range
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism in a desired operating position	PROPOSED	Is sensing modality specified; decision rules with (source data) preloaded
DO-017	Information path: sensing element -> decision/monitor	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle	UNKNOWN	Rig concept named, build specification UNKNOWN; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig concept	UNKNOWN	named, build specification UNKNOWN; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig concept	UNKNOWN	named, build specification UNKNOWN; acceptance instrumentation
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	Rig concept named, build specification UNKNOWN; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
scale deposition and fracture	Failure by scale deposition and fracture	Design deformation NOT ESSENTIAL (device failure signal unknown)	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode
kink/buckling below required	Failure by kink/buckling below required	compressions, buckling, etc. and suppression	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode
fatigue crack initiation	Failure by fatigue crack initiation	compressions, buckling, etc. and suppression	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode
wear debris generation	Failure by wear debris generation	compressions, buckling, etc. and suppression	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode
joint separation under error	Failure by joint separation under error	compressions, buckling, etc. and suppression	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode
set-release of super element	Failure by set-release of super element	compressions, buckling, etc. and suppression	ALVS-006 (device failure signal unknown)	UNKNOWN the invested candidate failure mode

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
scale deposition and pitch retention	Failure by material deposition and pitch deformation	designer control	NOT ESTABLISHED (defined after verification)
kink/buckling below required	Failure by kink/buckling below required	pushability test	pushability test
fatigue crack initiation at stress	Failure by fatigue crack initiation at stress	concentration of stress at surfaces	pushability test
wear debris generation at sliding	Failure by wear debris generation at sliding	paired and paired	pushability test
joint separation under overload	Failure by joint separation under overload	load on a larger size	pushability test
set-release of superelastic shape	Failure by set-release of superelastic shape	pushability test	pushability test

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Build one prototype with the pitch-retention	Build one prototype with the pitch-retention	Build one prototype with the pitch-retention	Build one prototype with the pitch-retention
VF-002	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty
VF-003	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance	fatigue life (run-out) testing. Acceptance
VF-004	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty	buckling/kink margin test across the duty
VF-005	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading
VF-006	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty	buckling / kink margin tests across duty
VF-007	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst	overload margin demonstration to worst

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite material	nitinol / polymer composite material	1	Material	nitinol / polymer composite material	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)
WP-02	article for VF-001	per method	Build one prototype with one verification capability	NOT ESTABLISHED (to be quoted)	one verification capability	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-002	per method	buckling / kink margin test	NOT ESTABLISHED (to be quoted)	VF-002 verification	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-003	per method	fatigue life (run-out) test	NOT ESTABLISHED (to be quoted)	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-004	per method	buckling/kink margin test	NOT ESTABLISHED (to be quoted)	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-005	per method	fatigue run-out at work	NOT ESTABLISHED (to be quoted)	VF-005 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-006	per method	buckling / kink margin test	NOT ESTABLISHED (to be quoted)	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-007	per method	overload margin demonstration	NOT ESTABLISHED (to be quoted)	VF-007 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 8-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.